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Hazomatic Sensor Build:

Files needed:

For the case construction: .stp file in the hardware folder
To flash to the ESP32 board: .bin file in the firmware folder
NB: The .bin file only works for ESP32C3 board specified, please get in touch if you have questions.

To buy:

SPS30 Sensirion sensor https://sensirion.com/company/news/press-releases-and-news/article/sensirion-inside-elv-pm25-particulate-matter-analyzer

I've been using Farnell as a supplier but lots of companies sell them.

An ESP32 C3 - for one that fits in the 3D printed enclosure and works with the .bin file, I've used this:

https://thepihut.com/products/baguette-c3-esp32-c3-prototyping-board?variant=54967506764161

Get the board without headers.

And to connect the sensor to the board:

https://thepihut.com/products/jst-zh-cable-5-pin-1-5mm-pitch

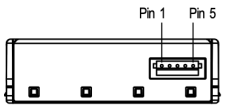
Again there are lots of suppliers for this, but you might as well get the PiHut one if you're ordering an esp32 from them.

A USB C power cable and power supply… The sensor does not need a beefy power supply.. probably whatever you have lying around! The sensor uses around 0.7 Watts whilst measuring data.

Construction

Plug in the connector to the sensor, then you need to solder 4 of the 5 cables to the board:

SPS30 Pin 1 5V to ESP32 5V
SPS30 Pin 2 Rx to ESP32 Tx
SPS30 Pin 3 Tx to ESP32 Rx
SPS30 Pin 4 NOT USED
SPS30 Pin 5 to ESP32 Ground



Pin	Name	Description	Comments
1	VDD	Supply voltage	5V ± 10%
2	RX	UART: Receiving pin for communication	TTL 5V and LVTTL 3.3V compatible
	SDA	I²C: Serial data input / output	
3	TX	UART: Transmitting pin for communication	TTL 5V and LVTTL 3.3V compatible
	SCL	I²C: Serial clock input	
4	SEL	Interface select	Leave floating to select UART
			Pull to GND to select I²C
5	GND	Ground	Housing on GND

Flash the Firmware

I've made flashing simpler.. It's a one time thing..

The firmware is the .bin file in the firmware folder
You will need a chrome based browser like Chrome, or Firefox - Safari will not work.
Connect the usb lead to the sensor, then whilst plugging the usb lead into your computer, hold down the bottom button on the sensor (you can release

once it's plugged in). This puts the sensor into flashing mode.

On your Chrome browser go to:
https://adafruit.github.io/Adafruit_WebSerial_ESPTool/

When you connect you will be asked which USB port your sensor is connected to on the computer.

Next load the supplied .bin file onto the web page in the first slot..

Press Program and let the tool do its thing…
(about 30 seconds)

Once programmed you can press the bottom button (closest to the usb port) to restart, or remove and add to power… You don't need to be connected to your computer anymore and flashing should be complete.

Set up Hazomatic

Power up Hazomatic sensor, the LED will briefly flash, and then light amber.

To allow the sensor to join your local wifi network you need to put the sensor into wifi discovery mode do the following:

Power the device, by pushing then releasing the button at the bottom of the unit to restart it. Then within the next 10 seconds, push and hold the top button until the led light turns blue. This indicates the sensor is in wifi set up mode.

On a computer, phone or tablet, look at available networks - you should see the hazomatic SSID - join the network and a splash screen should shortly appear on your device.

Go to set up

Set the device to a unique number to your setup (only important if you are using more than one sensor. This sets the mDNS (bonjour) name to something like hazomatic45.local (the 45 being your number).

Press Save and you should return to the home screen.

Select Configure WiFi

Select your Wifi network or enter the SSID manually.

Enter the wifi password for your lx network. Hazomatic sensor should reboot and join the network. When it restarts up the led will be amber, when it connects to the wifi network the led turns green.

(If there is a problem with the Hazomatic sensor and it is unable to take measurements the led turns red - check your wiring!).

Go to the app and the sensors tab to add the sensor to Hazomatic app.

You should also be able to connect to the sensor via a very simple web interface using either the mDNS/bonjour name of your sensor http://hazomatic45.local or the direct ip address.

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